

Richard Feng

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TECHNICAL SKILLS AND TOOLS

Languages: Python, Java, JavaScript, HTML & CSS, R, SQL
Frameworks & Libraries: React, Next.js, PyTorch, SpringBoot
Tools & Databases: GitLab, Docker, Jupyter, Maven, phpMyAdmin, MySQL, JDBC, PostgreSQL
Platforms & Development: Windows, macOS, Linux, Netlify, GCP, Godot, Supabase

EDUCATION

UW Madison (2022 – 2026): GPA: 3.762 / 4.00
Bachelor of Science (BS) in Computer Sciences
Honors Algorithms, Database Management Systems, Intro to AI, Computer Graphics
Big Data Systems, Software Engineering, Computer Networks, Capstone

Bachelor of Science (BS) in Data Science
R Data Science Modeling II, Nonparametric Statistics, BioMed Image Analysis

Certificate in Mathematics
Advanced Multivar, Advanced Linear, Cryptography, Combinatorics

EXPERIENCE

Seeds of Success – Software Developer II January 2026 – Current

- Serve as the primary point of contact between Seeds of Success and Invisible Children's (KARA) development team, coordinating directly with their lead developer to align on technical decisions and accelerate full-stack feature delivery
- Collaborated with KARA's lead developer on frontend architecture decisions including component structure and routing to ensure a scalable, maintainable codebase
- Built full-stack research document retrieval features using OpenSearch, including API endpoints and a Jupyter notebook integration for efficient data access by policymakers and researchers
- Developed internal context-management systems to maintain live organizational knowledge, reducing LLM token usage and improving response quality

Senra Systems – Computer Science Capstone February 2026 – April 2026

- Designed and built a tool that automates initial workflows for wire harness manufacturing by extracting an iterable graph of connectors from visual schematics and utilizing RAGs for preliminary design review and manufacture instructions
- Architected and implemented a PostgreSQL database to support persistent data storage across the application, including schema design and integration with backend services and RAG data retrieval
- Built full stack features end-to-end, including user authentication and persistent project saving
- Turned 3 days of work into 30 minutes at ~\$0.25 per harness table and ~\$5 per Engineering Master with 40% part spec accuracy

PROJECTS

OCT Retinal Disease Classifier April 2026 – May 2026

- Built deep learning pipeline achieving 95% accuracy in classifying retinal diseases from 109,000+ OCT images across 4 categories (CNV, DME, DRUSEN, Normal)
- Designed preprocessing pipeline with multi-scale Gaussian denoising and CLAHE contrast enhancement, improving DRUSEN detection precision by 6%
- Implemented class-weighted loss functions to handle 5:1 class imbalance, achieving clinical-grade performance (0.82 F1-score) on rare pathologies
- Authored research paper analyzing architecture depth vs. preprocessing trade-offs; published code and documentation on GitHub

Computer Networks: Virtual Switch, Router & TCP October 2025 – December 2025

- Implemented a learning Ethernet switch with MAC-based forwarding and timeout aging in Java using Mininet and POX
- Built a virtual IP router with longest-prefix-match route lookups, IPv4 checksum verification, TTL decrement, and ARP caching
- Extended the router with RIPv2 distance-vector routing, including heartbeats, triggered responses, and route expiry for dynamic topology convergence
- Implemented TCP Tahoe over UDP, including slow start, congestion avoidance, and fast retransmit

Asteroids February 2025 – April 2025

- Collaborated in a team of 6 to build a full-stack Asteroids game with user accounts, stat tracking, and leaderboards
- Defined the full tech stack (React, REST, Spring Boot, JDBC, MySQL) and contributed to cross-layer architecture decisions
- Designed and implemented the gameplay interface in React and integrated real-time game state via REST APIs